DISCUSSION PAPER SERIES

DP19768

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MONETARY ECONOMICS AND FLUCTUATIONS

ISSN 0265-8003

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Discussion Paper DP19768
Published 14 December 2024
Submitted 06 December 2024

Centre for Economic Policy Research
33 Great Sutton Street, London EC1V 0DX, UK
187 boulevard Saint-Germain, 75007 Paris, France
Tel: +44 (0)20 7183 8801
www.cepr.org

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Abstract

Sims (2013) intended to illustrate the Fiscal Theory of the Price Level (FTPL).
This comment shows that in his endeavor, Sims (2013) overlooked the fact that the FTPL does not extend to overlapping generations (OLG) or to incomplete markets models. Otherwise he would have recognized that his simplifying assumptions are indeed knife-edge. Furthermore Brunnermeier et al. (2023) would not have based their misunderstanding of the workings of the FTPL in OLG and incomplete markets models on Sims (2013). Likewise Bassetto and Sargent's (2020) claim, that it is sufficient for fiscal policy to implicitly provide a nominal anchor through setting the real primary surplus, does not extend to OLG or incomplete markets models. To ensure determinacy, fiscal policy has to explicitly set a nominal anchor as envisaged by the seminal work of Hagedorn (2016).

JEL Classification:

Keywords:

Marcus Hagedorn - marcus.hagedorn07@gmail.com
University of Oslo and CEPR

A Comment on "Paper Money" by Chris Sims*

Marcus Hagedorn†
University of Oslo and CEPR

This Version: November 18, 2024

Abstract

Sims (2013) intended to illustrate the Fiscal Theory of the Price Level (FTPL). This comment shows that in his endeavor, Sims (2013) overlooked the fact that the FTPL does not extend to overlapping generations (OLG) or to incomplete markets models. Otherwise he would have recognized that his simplifying assumptions are indeed knife-edge. Furthermore Brunnermeier et al. (2023) would not have based their misunderstanding of the workings of the FTPL in OLG and incomplete markets models on Sims (2013). Likewise Bassetto and Sargent's (2020) claim, that it is sufficient for fiscal policy to *implicitly* provide a nominal anchor through setting the real primary surplus, does not extend to OLG or incomplete markets models. To ensure determinacy, fiscal policy has to *explicitly* set a nominal anchor as envisaged by the seminal work of Hagedorn (2016).

JEL Classification: D52, E31, E43, E52, E62, E63
Keywords: FTPL, Fiscal and Monetary Policy, Inflation, Ricardian Equivalence, Price level determinacy

*Support from the FRIPRO Grant No. 250617 is gratefully acknowledged. I thank Morten Ravn for encouraging me to write this paper.

†University of Oslo, Department of Economics, Box 1095 Blindern, 0317 Oslo, Norway.
Email: marcus.hagedorn@econ.uio.no

Sims (2013) uses a two-period overlapping-generations model to argue that imposing a zero tax delivers indeterminacy and that instead imposing a (small) positive tax, delivers a unique equilibrium. The conclusion is that tax backing renders an indeterminate price level determinate, that is, the FTPL works. The present comment shows that this conclusion hinges on knife-edge assumptions (zero old age endowment and nonnegative savings) and that without these assumptions we obtain steady state multiplicity, demonstrating that Sims (2013) overlooked the fact that the FTPL does not extend to OLG models. Accordingly, the OLG model cannot be used to illustrate the FTPL as intended by Sims (2013, Page 564), but instead reveals its limitations.

Sims (2013) also considers a representative agent model in order to combine an interest rate rule satisfying the Taylor principle with a fiscal rule, where the primary surplus responds to inflation [Section 3B: Fiscal Backup for a Taylor Rule]. I again show that imposing these monetary and fiscal rules in an OLG model delivers indeterminacy, that is, the FTPL does not extend to OLG models.

The deeper reason for the multiplicity as explained in Hagedorn (2024), is that inflation and real interest rates are not determined through the FTPL mechanism. The implication is that multiple steady states with different real interest rates can indeed exist. Whether multiple steady states actually exist depends on other assumptions on the environment unrelated to the FTPL. This comment shows that these assumptions are knife-edge. Sims (2013) overlooked this and consequently also overlooked that the FTPL does not extend to OLG models. In particular Sims (2013) misses the connection between the FTPL and the well-known finding that OLG models can feature multiple steady states. Footnote 1 in Sims (2013) also points in this direction,

> "Many of the insights of the FTPL literature are implicit in Neil Wallace's earlier paper (1981). Wallace presented a model in which conventional monetary policy actions had no effect on the price level so long as fiscal policy were held constant. Of course this implied that in his model fiscal policy determined the price level, though Wallace did not put it that way."

Irrelevance means that the sequence of open market operations supporting *a given* equilibrium is not unique. Irrelevance does not "of course" imply that the equilibrium *is* unique and in particular that fiscal policy ensures uniqueness of an equilibrium, what Sims (2013) seems to assume.[1]

[1]Perhaps "Wallace did not put in that way", since he emphasized that "The irrelevance proposition

Brunnermeier et al. (2023) accept that the FTPL also works in OLG models without any further own scrutiny, merely based on the fact that Sims (2013) is ”written by a well-respected authority in the field“ [verbatim!]. Similarly, despite the very demanding referee process at Econometrica, Angeletos et al. (2024a) overlook the multiplicity, rendering their theoretical results incorrect.[2] This comment points out and clarifies this very persistent and important misconception in the literature.

At a more conceptual level, the reason for the indeterminacy as envisaged by Hagedorn (2016) is that steady-state uniqueness requires fiscal policy to explicitly set a nominal anchor, which determines the inflation rate and which in turn determines, together with the nominal interest rate set by monetary policy, the real interest rate. Instead, Bassetto and Sargent (2020) recognized correctly that a nominal anchor is needed, but that the FTPL, if it operates, only provides an indirect anchor, in the sense that a unique price level path delivers a nominal anchor. However, this conclusion does not extend to OLG or incomplete markets models. In this model class, the FTPL does not ensure price-level determinacy and as a result does not deliver a nominal anchor. Instead, price-level determiancy requires a direct nominal anchor provided by fiscal policy as envisaged by Hagedorn (2016).

To further illustrate the deeper problem for multiplicity when applying the FTPL, I use the classic paper of Benhabib et al. (2002). They find that an interest rate rule following the Taylor principle, and taking into account the zero(effective) lower bound on nominal interest rate, delivers two steady states with different inflation rates. I show that the FTPL cannot overcome this multiplicity, simply because it does not determine the inflation rate. Instead, uniqueness is restored if fiscal policy provides a nominal anchor (Hagedorn, 2016).

Sims is right that “our thinking and teaching about inflation, monetary policy, and fiscal policy should be based on models that recognize fiscal-monetary policy interactions.” At the same time, our thinking should recognize that markets are incomplete. Furthermore, incomplete markets imply this fiscal-monetary policy interaction as described in the Demand Theory of the Price Level (DTPL) developped in Hagedorn (2016), simply because the DTPL, but not the FTPL, operates in incomplete markets models.

applies to asset exchanges under some conditions” and that variation in money supplies can alter the price level in the usual way.

[2]The errors in Angeletos et al. (2024a) carry over to Angeletos et al. (2024b) and are exacerbated by the authors’ failure to obey the rules of good scientific practice.

1 The OLG Model in Sims (2013)

I briefly describe the relevant aspects of the overlapping generations (OLG) model in Sims (2013), using (almost) the same notation in order to enhance readability. Sims (2013) starts with a model with storage, and then shows that "By imposing the tax, no matter how small, the government has eliminated all those equilibria in which storage and debt coexist." I therefore describe the model without storage from the outset, since the main focus of this comment is on this model. I then return to the model with storage and the incorrect intuition in Sims (2013) when explaining my findings for the OLG model without storage.

Time is discrete. At each point in time t, two generations of measure one, (y)oung and (o)ld, are alive, and each generation lives for two periods. The young generation born in Period t derives utility from consumption when young, C_{1t}, and consumption when old, $C_{2,t+1}$,

$$U(C_{1t}, C_{2,t+1}). \tag{1}$$

Period t young households have a constant real endowment ω^y normalized to one in Sims (2013), pay real taxes τ, are born without any assets, can acquire nominal bonds B_t, such that their budget constraint reads

$$C_{1t} + \tau + \frac{B_t}{P_t} = \omega_y = 1 \tag{2}$$

where P_t is the period t price level and τ is a lump-sum tax. Period $t+1$ old households have a constant real endowment ω^o and real asset income $\frac{R_t B_t}{P_{t+1}}$ for a gross nominal interest rate $R_t = 1 + i_t$, such that their budget constraint equals

$$C_{2,t+1} = \omega^o + \frac{R_t B_t}{P_{t+1}}. \tag{3}$$

Sims (2013) assumes that R is constant in the OLG model, sets old age endowment to zero, $\omega^o = 0$, and consequently imposes a constraint $B_t \geq 0$ to ensure that old age consumption is not negative. I show that this is the critical assumption in Sims (2013), and that instead assuming $\omega^o > 0$ and consequently dispensing with $B_t \geq 0$, overturns the Sims's (2013) conclusion on the FTPL.

The government imposes a lump-sum tax τ and issues nominal debt D_t, such that the government budget constraint reads

$$\frac{D_t}{P_t} = (1 + r_{t-1})\frac{D_{t-1}}{P_{t-1}} - \tau,$$

where $1 + r_t = \frac{R_t P_t}{P_{t+1}}$ is the gross real interest rate.

Equilibrium

An equilibrium is a sequence of prices and allocations such that:

(i) households maximize utility, given prices P_t, R_t and taxes τ, subject to the budget constraints (2) and (3),

(ii) the government flow budget constraint is satisfied,

(iii) the resource constraint is satisfied:

$$C_{1t} + C_{2,t} = \omega_y + \omega_o,$$

(iv) the asset market clears: $D_t = B_t$.

I now first derive the household saving function without imposing the assumptions in Sims (2013), and then show how the specific assumptions in Sims (2013) generate his result, and that relaxing them leads to the opposite conclusion.

Household Aggregate Saving

Sims (2013) assumes $U(C_{1t}, C_{2,t+1}) = log(C_{1t}) + log(C_{2,t+1})$. I generalize the utility function and assume for $\sigma > 0$

$$U(C_{1t}, C_{2,t+1}) = \frac{C_{1t}^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} + \frac{C_{2,t+1}^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}, \tag{4}$$

which nests the utility function in Sims (2013) for $\sigma = 1$.

I define real household savings, $b_t := \frac{B_t}{P_t}$. Young household optimization leads to the first-order condition

$$[\omega^y - \tau - b_t]^{\frac{-1}{\sigma}} = \beta(1 + r_t)[(\omega^o + (1 + r_t)b_t]^{\frac{-1}{\sigma}}. \tag{5}$$

Solving for real savings b_t,

$$b_t = \frac{\omega^y - \tau - [\beta(1 + r_t)]^{-\sigma}\omega^o}{1 + \beta^{-\sigma}(1 + r_t)^{1-\sigma}}. \tag{6}$$

Imposing a steady state with a constant real interest rate, r_{ss}, yields steady-state savings

$$b_{ss}(r_{ss}) = \frac{\omega^y - \tau - [\beta(1+r_{ss})]^{-\sigma}\omega^o}{1+\beta^{-\sigma}(1+r_{ss})^{1-\sigma}}. \tag{7}$$

Denoting real government debt, $d_t = \frac{D_t}{P_t}$, yields in steady state

$$r_{ss}d_{ss} = \tau. \tag{8}$$

Thus a steady state exists iff $d_{ss} = b_{ss}$. If $r \neq 0$ (which cannot be an equilibrium in the relevant case $\tau > 0$ considered in Sims (2013)), this is equivalent to

$$b_{ss}(r_{ss}) = \frac{\omega^y - \tau - [\beta(1+r_{ss})]^{-\sigma}\omega^o}{1+\beta^{-\sigma}(1+r_{ss})^{1-\sigma}} = \frac{\tau}{r_{ss}} = d_{ss}. \tag{9}$$

A steady-state interest rate r_{ss} thus satisfies:

$$\Gamma(r) := r\frac{\omega^y - \tau - [\beta(1+r)]^{-\sigma}\omega^o}{1+\beta^{-\sigma}(1+r)^{1-\sigma}} - \tau = 0. \tag{10}$$

If old age endowment is positive, $\omega^o > 0$,

$$\lim_{r\to-1}\Gamma(r) = \infty, \qquad \lim_{r\to\infty}\Gamma(r) = \infty. \tag{11}$$

for all $\sigma > 0$. Since $\Gamma(r = 0) < 0$, this implies that two steady states exist, one with real interest $r_1 < 0$ and one with real interest $r_2 > 0$.

Result 1. *If old-age income is positive (and without exogenous restrictions on savings B) two steady states with real interest rate $r_1 < 0$ and $r_2 > 0$ exist. Setting a fixed positive surplus $\tau > 0$ thus does not ensure uniqueness, that is, the FTPL does not extend to OLG models.*

The claim in Sims (2013, page 574) "... while small positive values of τ make equilibrium unique" is correct under his knife-edge assumptions, but it is incorrect if the old generation has positive endowment and consequently, young households can have negative savings. If $\omega^o > 0$, then imposing a (small) tax $\tau > 0$ does not ensure uniqueness. Uniqueness can only be ensured, if by assumption, the $r_1 < 0$ equilibrium is ruled out, e.g. by imposing $B \geq 0$.[3] I will now show that small (standard) modifications of the model imply that one

[3] Similarly in Sims (2013, page 573) "By imposing the tax, no matter how small, the government has eliminated all those equilibria in which storage and debt coexist." relies on the assumptions $\omega^o > 0$ and $B \geq 0$. Without imposing these assumptions, storage and debt *can* coexist.

obtains multiple steady states even if $B \geq 0$ is imposed.

Multiplicity: Taylor principle and ZLB

Assume a standard interest rate rule satisfying the "Taylor principle" and taking into account the ZLB,

$$R_t = \max\{1, (1 + \pi_t)^{\phi}\} \tag{12}$$

with $\phi > 1$ and the inflation rate $1 + \pi_t = \frac{P_t}{P_{t-1}}$.

Consider the positive equilibrium real interest rate $1 + r_2$ from Result 1. Set $\pi_a < 0$ such that

$$1 + r_2 = \frac{1}{1 + \pi_a},$$

and $\pi_b > 0$ such that

$$1 + r_2 = (1 + \pi_b)^{\phi - 1}.$$

Both inflation rates π_a and π_b are steady-state, which yield the same real interest rate, $1 + r_2$. The difference is that the ZLB is binding for π_a and not for π_b. This finding is a slight adaptation of Benhabib et al. (2002), as it uses the idea that the ZLB implies two different steady-state inflation rates with the same real interest rate. The interest rate rule (12) eliminates the $r_1 < 0$ steady state, but delivers a new steady state. Making ad-hoc assumptions like $B \geq 0$ does not overcome the multiplicity.

Result 2. *If monetary policy follows an interest rate rule (12), then there are two steady states with positive real interest. Setting a fixed positive surplus $\tau > 0$ thus does not ensure uniqueness, even if $B \geq 0$ is exogenously imposed, that is, the FTPL again does not extend to OLG models.*

Multiplicity in Sims' Section 3B: Taylor principle without ZLB

The deeper reason as explained in Hagedorn (2024), is that the FTPL does not determine the steady-state inflation rate. For this reason, the analysis in Sims (2013, Section 3B), which assumes a Taylor interest rate rule without imposing the ZLB, and a fiscal rule $\tau_t = \bar{\tau} + \kappa \pi_t$ for $\kappa > 0$, does not carry over to an OLG model, even if $B \geq 0$ or $\omega^o = 0$ is assumed and no ZLB is imposed. For a standard inflation coefficient $\phi = 1.5$, generically there is either no steady state or (at least) two steady states. To see this, note that the steady-state government budget constraint reads

$$r_{ss}d_{ss} = \bar{\tau} + \kappa\pi_{ss} = \bar{\tau} + \kappa((1+r_{ss})^{\frac{1}{\phi-1}} - 1)$$

If $r \neq 0$, asset market clearing is equivalent to[4]

$$\Gamma(r) = r\frac{\omega^y - \tau - [\beta(1+r)]^{-\sigma}\omega^o}{1+\beta^{-\sigma}(1+r)^{1-\sigma}} - [\bar{\tau} + \kappa((1+r_{ss})^{\frac{1}{\phi-1}} - 1)] = 0. \quad (13)$$

Since $\lim_{r\to 0}\Gamma(r) = -\bar{\tau} < 0$ and $\lim_{r\to\infty}\Gamma(r) = -\infty$, there is no steady state if $\Gamma(r) < 0$ for all $r \geq 0$ and at least two steady states if an $\hat{r}$ exists with $\Gamma(\hat{r}) > 0$.[5]

Result 3. *If monetary policy follows an interest rate rule $R_t = (1+\pi_t)^{1.5}$, and fiscal policy is described by the tax rule $\tau_t = \bar{\tau} + \kappa\pi_t$ with $\bar{\tau}, \kappa > 0$, then generically, either zero or two steady states with positive real interests exist. Setting an exogenous tax rule thus does not ensure uniqueness, even if $B \geq 0$ is exogenously imposed, that is, the FTPL again does not extend to OLG models.*

Three-generations model with multiple positive steady-state real interest rates

Considering a three-, instead of a two-period, overlapping generations model is another way to obtain multiplicity even if $B \geq 0$ is imposed (Hagedorn, 2024).

Figure 1 provides an example of a three-generations model in which the aggregate savings function and the budget function $d_{ss} = \frac{\tau}{r_{ss}}$ intersect three times at a positive real interest, implying that three different price levels constitute an equilibrium, each associated with a different real interest rate and a different positive real value of bonds.[6] For expositional reasons, I do not show the third steady state equilibrium which arises for a high real interest rate.[7]

Again, the multiplicity arises, since the FTPL does not provide a theory of real interest or inflation rate determination.

[4]Note that the government budget constraint implies that $r_{ss} > 0$ if $B \geq 0$ is (exogenously) imposed.

[5]Note that $\Gamma(r) \leq 0 \ \forall r$ and $\Gamma(\hat{r}) = 0$ for one $\hat{r}$ does not arise generically.

[6]There are three generations of measure one with instantaneous utility function $\frac{c^{1-\sigma}}{1-\sigma}$ and $\sigma = 5$. The first generation has endowment 1 and pays taxes $\tau = 0.05$, the second generation has endowment 8 and pays no taxes, and the third generation has no endowment and pays no taxes. The discount factor is set to 1.

[7]Farmer and Zabczyk (2022) establish that their calibrated 62-generation model features much richer real indeterminacies than my simple model.

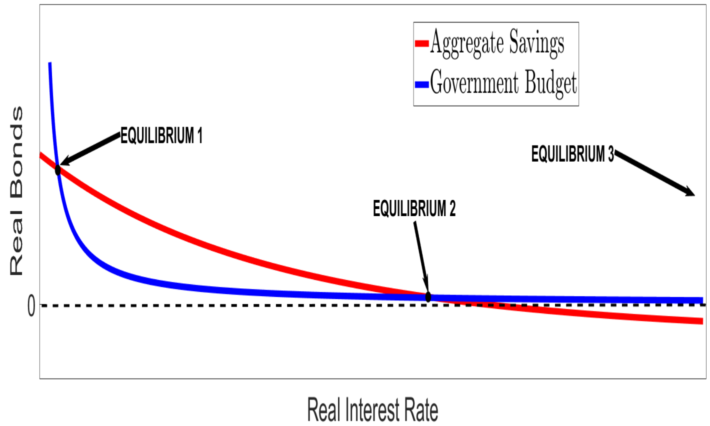

Figure 1: Three-generation OLG model with multiple equilibrium price levels

Note - The aggregate savings function is downward sloping and intersects the downward sloping government budget constraint $d_{ss} = \tau/r$ twice in the $(\frac{B}{P}, 1+r)$ space. A third intersection is not shown. Each intersection corresponds to a different equilibrium. Each equilibrium features a different price level, in contrast to the FTPL, which predicts a unique price level.

Result 4. *Multiple steady states with positive real interests can exist in three-generations models. Setting a fixed positive surplus $\tau > 0$ thus does ensure uniqueness, even if $B \geq 0$ is exogenously imposed, that is, the FTPL again does not extend to such OLG models.*

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